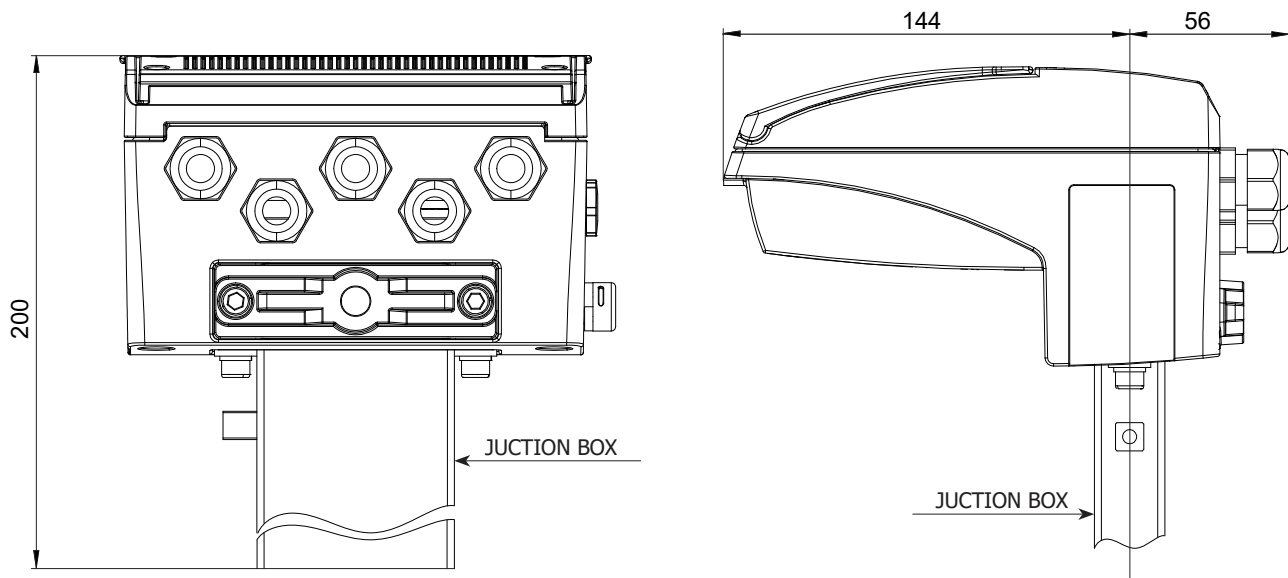
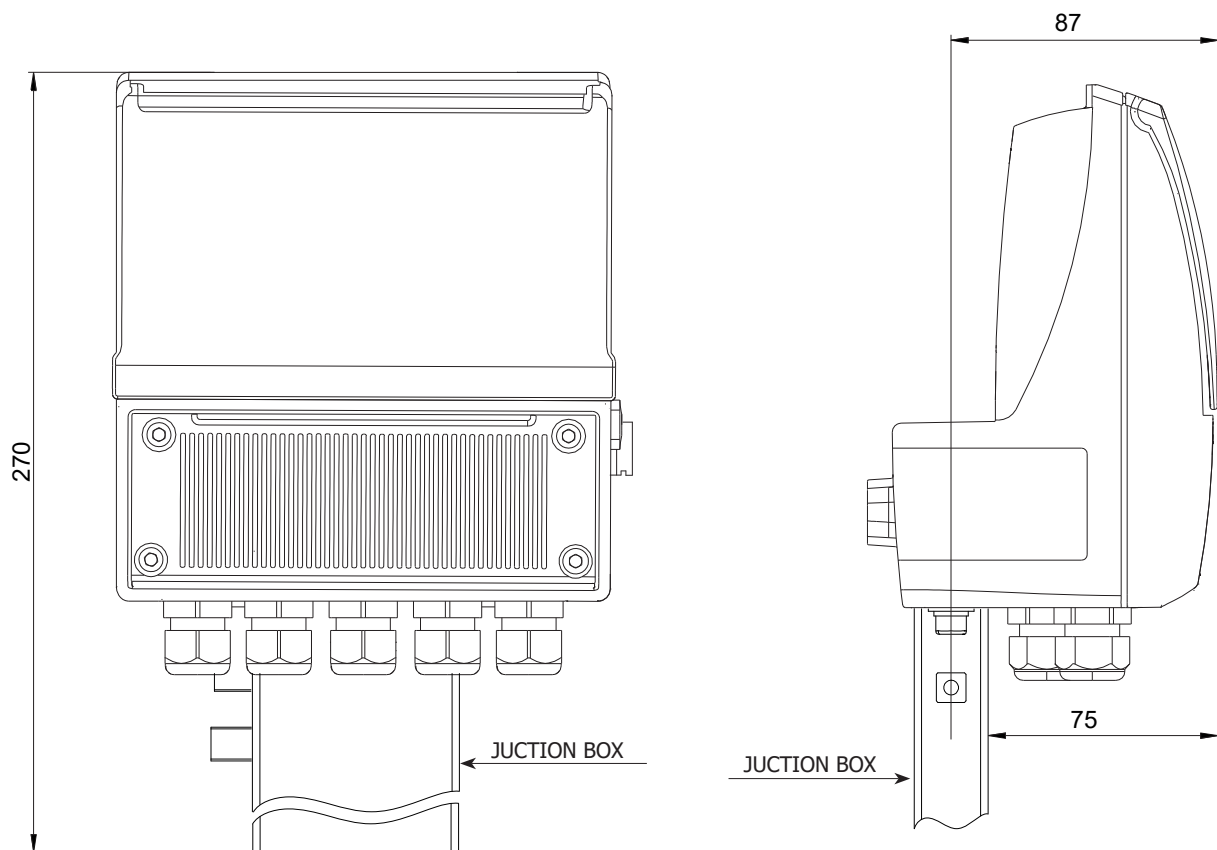


HORIZONTAL VERSION



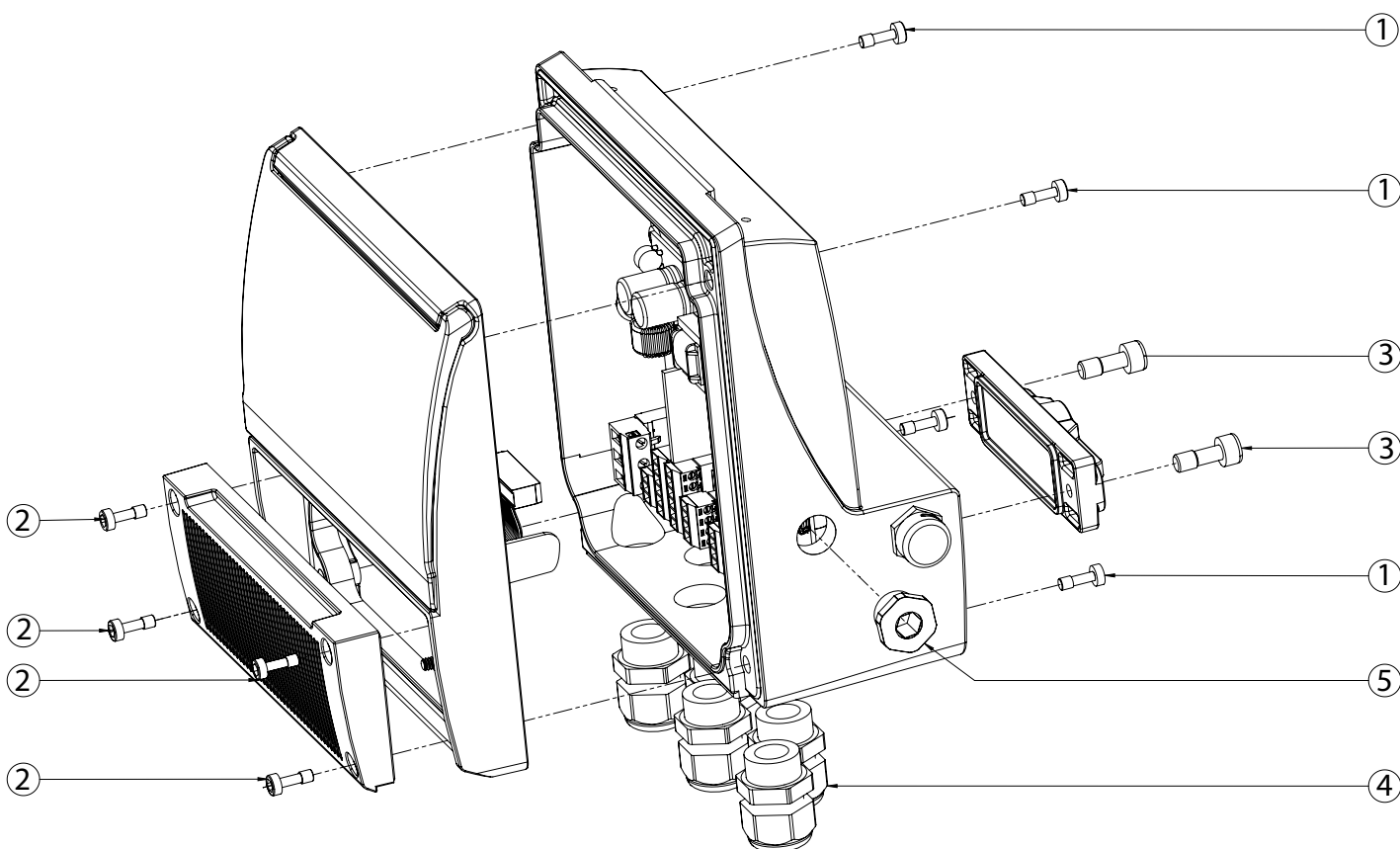
VERTICAL VERSION



TORQUES

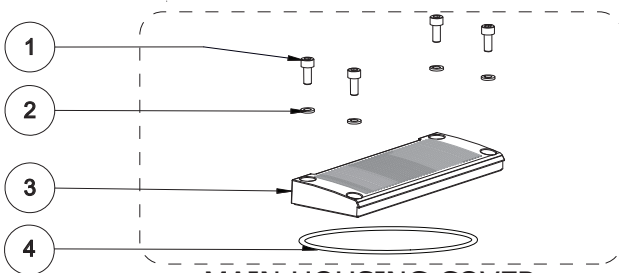
To guarantee the housing's IP degree the following torques are required:

HOUSING MATERIAL	Housing screws (1)	Screws Terminal block cover (2)	Fixing Display Frame	PCB Screws	Version Cap (3)	Cable Glands (4)	Cap USB-B (5)
ALUMINIUM HOUSING	6 Nm	5.5 Nm	3 Nm	0.8 Nm	8 Nm	4 Nm	4 Nm
PLASTIC HOUSING	2 Nm	2 Nm	2.5 Nm	0.8 Nm	7 Nm	4 Nm	4 Nm

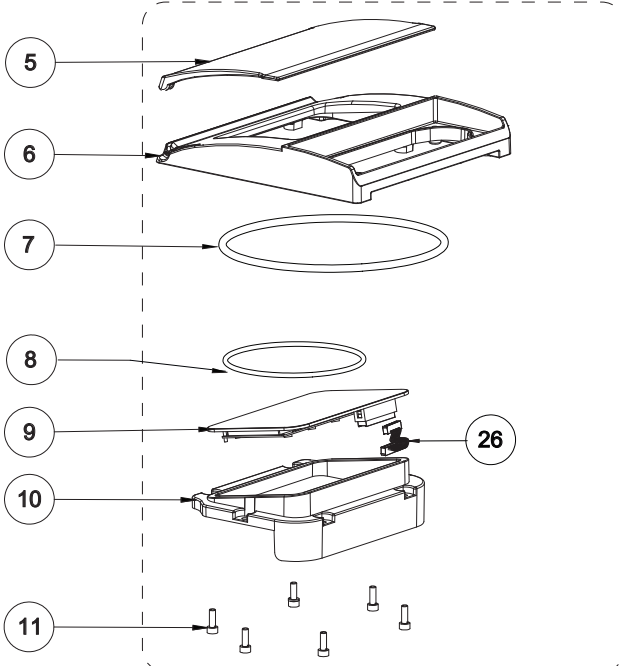


MV210 CONSTRUCTION

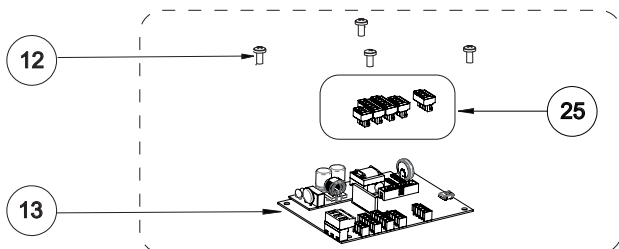
TERMINAL BLOCK COVER



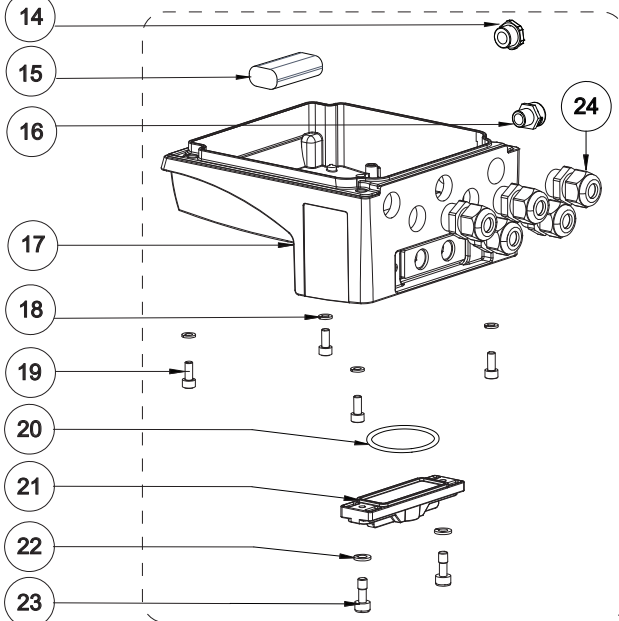
MAIN HOUSING COVER



PCB MV210



MAIN HOUSING

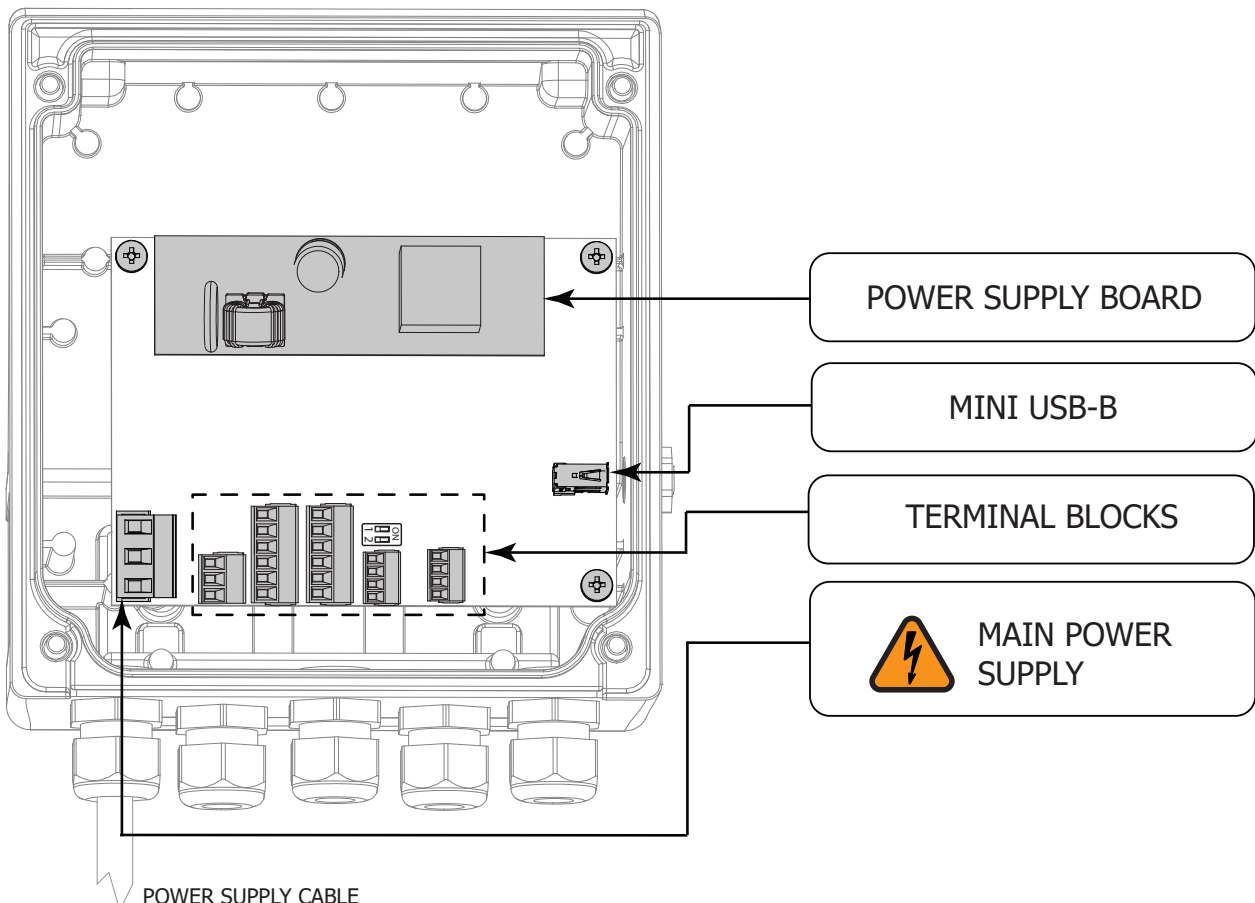
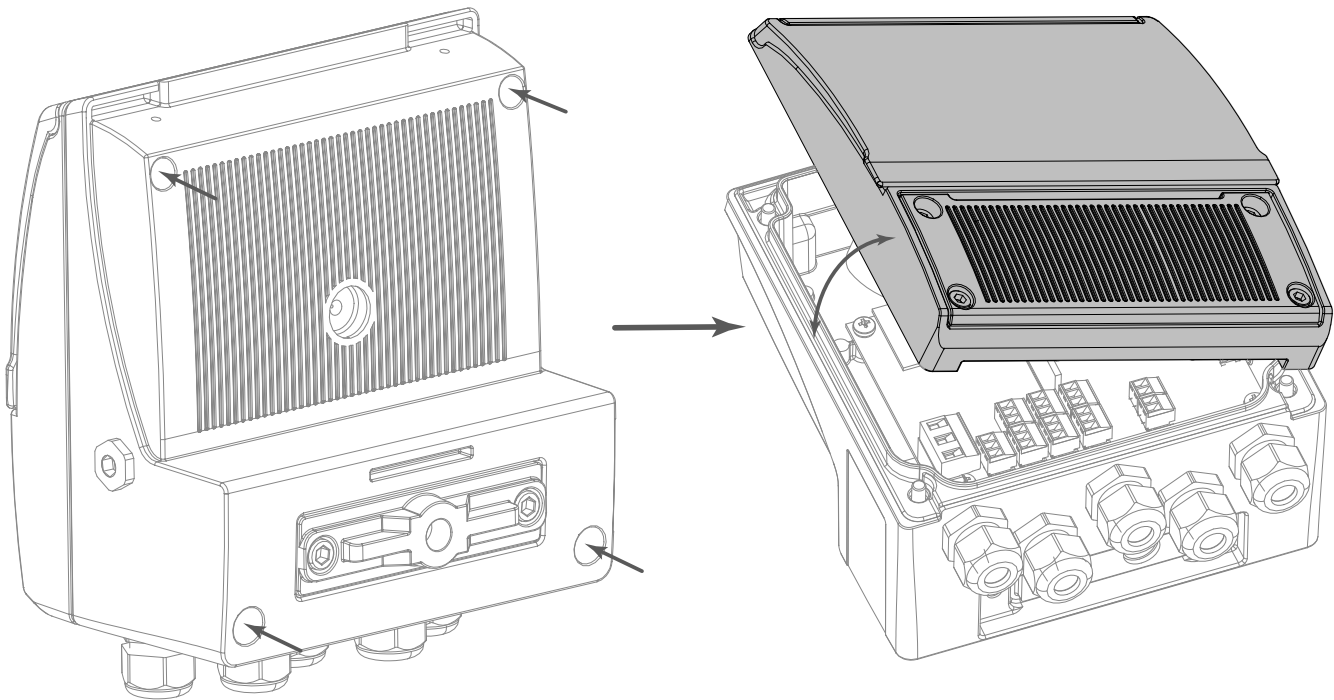


POS.	DESCRIPTION	
	PA6 VERSION	ALUMINIUM VERSION
1	SCREW M4x12	SCREW M5x12
2	GROWER Ø4	GROWER Ø5
3	TERMINAL BLOCK COVER	TERMINAL BLOCK COVER
4	O-RING-4400	
5	PROTECTION COVER	
6	HOUSING COVER	HOUSING COVER
7	ORING-4700	
8	ORING-117x3	
9	DISPLAY	
10	FIXING DISPLAY FRAME (MATERIAL PA06)	
11	SELF-TAPPING SCREW 4x10	TRILOBO SCREW 4x10
12	SELF-TAPPING SCREW 4x10	TRILOBO SCREW 4x10
13	PCB MV210	
14	PG9 CAP	
15	LITHIUM BATTERY	
16	ANTICONSEDE CAP	
17	PA6 MAIN HOUSING	ALUMINIUM MAIN HOUSING
18	GROWER Ø4	GROWER Ø5
19	SCREW M4x12	SCREW M5x12
20	O-RING-155	
21	VERSION CAP (MATERIAL PA06)	
22	SCREW M6x16	
23	GROWER Ø6	
24	PG11 CABLE GLAND CABLE DIAMETER: Ø5-Ø10mm	
25	TERMINAL BLOCK SOLID WIRE: 26-16 AWG / 0.129-1.31 mm² STRANDED WIRE: 26-16 AWG / 0.129-1.31 mm² TORQUE: 3.0 Lb.In / 0.34 Nm	
26	FLAT CABLE	

INTERNAL LAYOUT

INTERNAL CONVERTER VIEWS

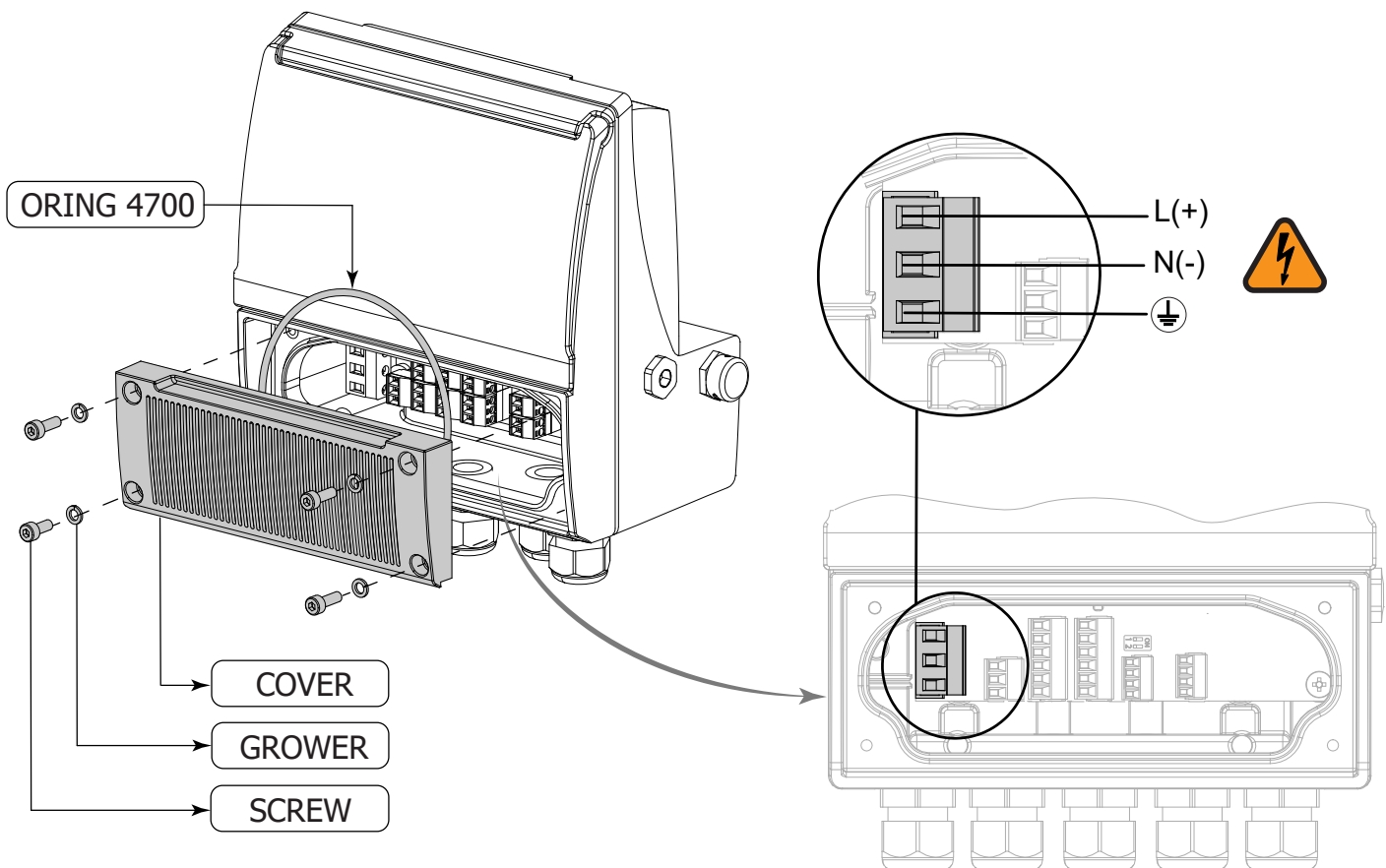
Remove the main housing cover by removing the 4 screws as shown here below.



ELECTRICAL CONNECTION AND GROUNDING INSTRUCTIONS



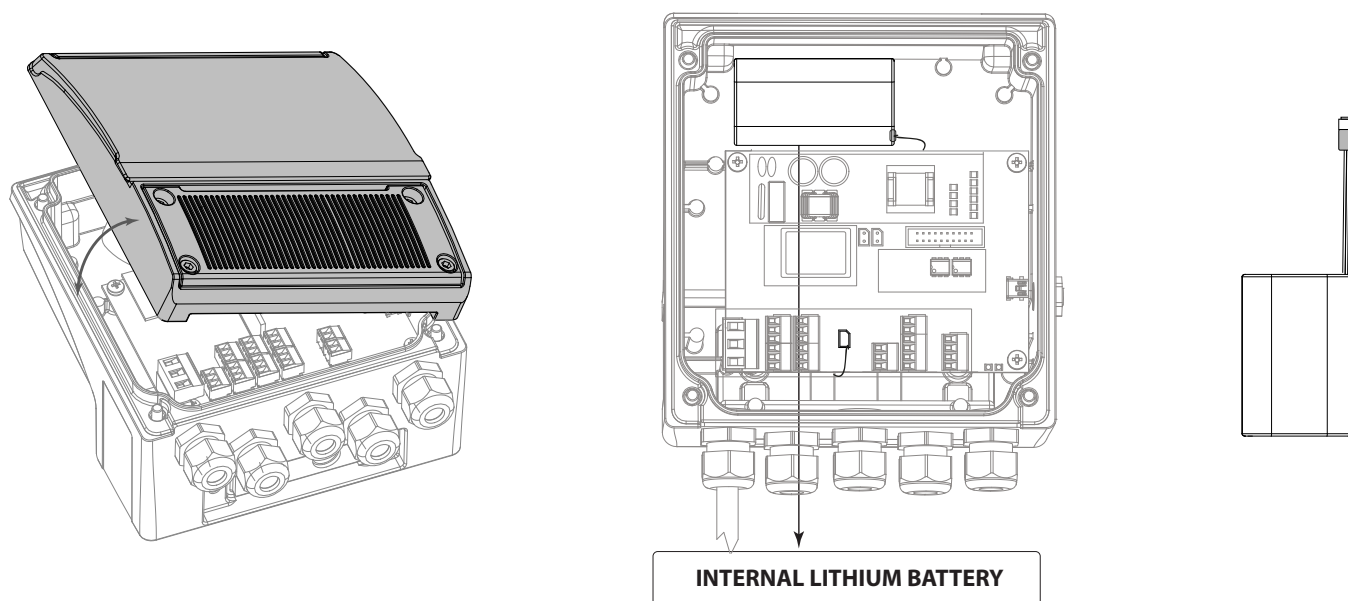
Always ensure that the converter and the sensor are grounded (earthed) correctly. The grounding of the sensor and converter **must** ensure that the instrument and liquid are equipotential.



- ❑ Before connecting the power supply, verify that the mains voltage is within the limits indicated on data plate.
- ❑ For the connections use only approved conductors, with fire-proof properties, whose section varies from 0.25mm² to 1.50mm², based on distance/power; additionally fix the power supply wires with a additional fastening system located close to the terminal.
- ❑ The power supply line must be equipped with an external protection for overload current (fuse or automatic line breaker).
- ❑ Provide in close proximity the converter a circuit breaker easily accessible for the operator and clearly identified; whose symbols must conform to the electrical safety and local electrical requirements.
- ❑ Ensure that the component complies with the requirements of the standard for electrical safety distance.
- ❑ Check chemical compatibility of materials used in the connection security systems in order to minimize electrochemical corrosion. In the aluminum housing it should avoid direct contact between the ground connection cable and the aluminum housing. It is therefore recommended to connect the safety ground cable, by placing it between the washer and the metal bracket on the related terminal or use an eyelet terminal crimped on the ground protection cable.
- ❑ The sensor, hardwired inputs and outputs are connected to the converter through terminal blocks located inside the converter.
- ❑ To locate the terminal block loosen the 4 screws on the terminal block cover. When the front cover is lifted, the terminal block is visible. The terminal block is the hardwire connection of the converter to external equipment, including the sensor.

The following pages give informations on the terminal block numbering, and the respective connecting of the sensor cables, and inputs/outputs.

INTERNAL LITHIUM BATTERY



- ❑ The internal lithium battery is rechargeable and should never be disconnected from the converter card when it is powered by the mains voltage. If this operation should be carried out it could irreparably damage the converter board.
- ❑ The internal lithium battery is recharged only when the converter is connected to the mains supply (LV, LLV, HV), battery charging condition, and not with the USB connection.
- ❑ During charging, the battery symbol appears on the MCP display and flashes blue; see "MEANING OF FLAGS" page 21 (the colors of the symbols can only be viewed in the virtual display of the MCP interface)
- ❑ When the battery charge falls below the minimum potential, battery low, the fixed red battery symbol appears; see "MEANING OF FLAGS" page 21 (the symbol colors can only be displayed in the virtual display of the MCP interface). Furthermore, in this charging condition below the minimum potential, the measurement does not start when the USB cable is connected.
- ❑ The thresholds that identify the condition of Battery low and battery charging are established by the system according to the use and settings assigned and therefore there is no fixed value..
- ❑ The two Battery low and battery charging icons can both be present as each one indicates a different condition.

GENERAL OPERATING NOTES

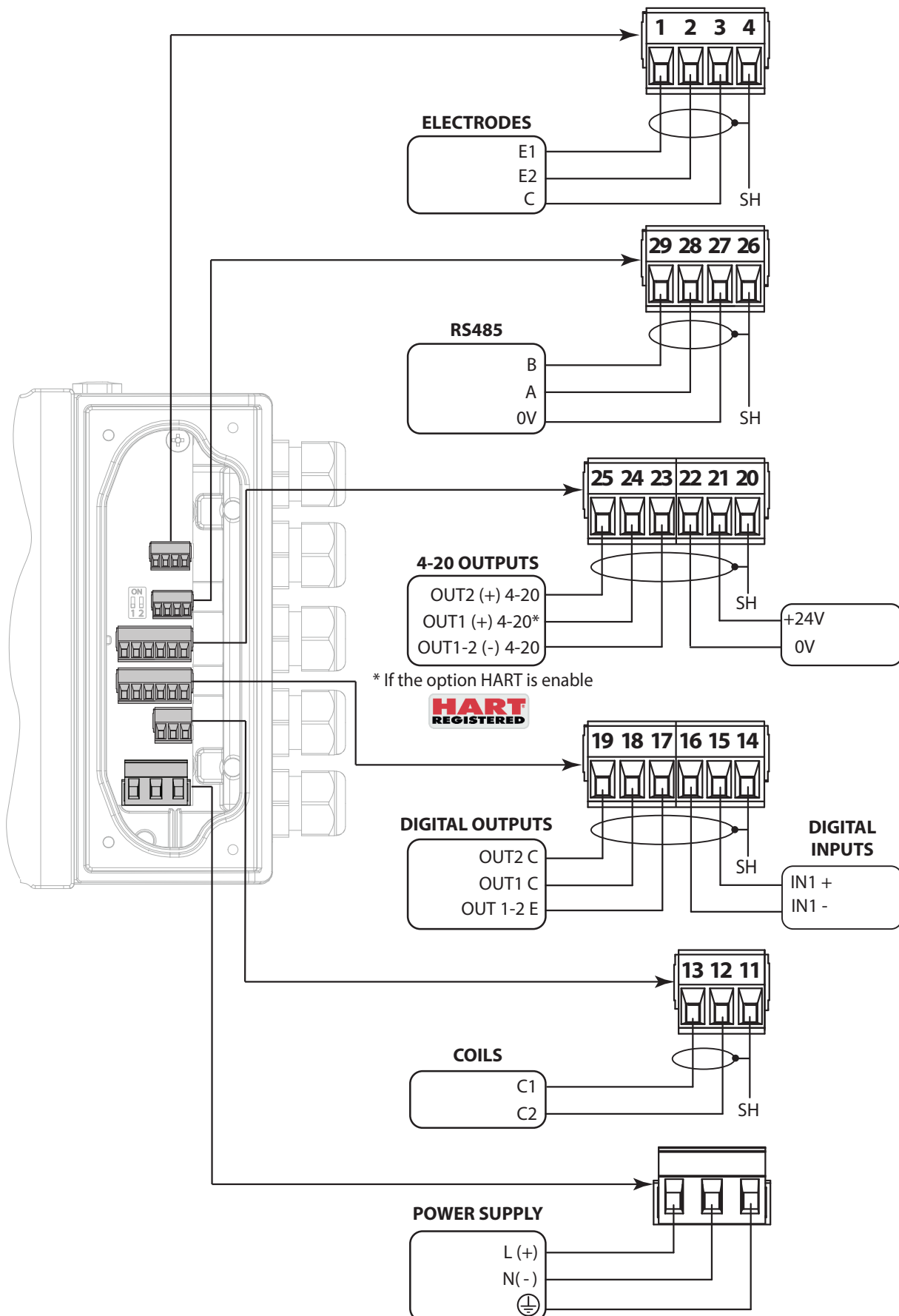
- ❑ If the mains voltage is disconnected from the converter with HV power supply, the flashing blue icon indicating the charging status can remain active for tens of seconds. This is due to the energy stored in the HV power supply and in the capacities of the circuit that discharge slowly.
- ❑ The battery voltage during charging does not rise immediately but gradually. This is related to the parameters set in the converter and detected by the controller chip.
- ❑ When the battery is physically disconnected the potential measured by the system is not true because the charging circuit checks the battery status by emitting impulses. These load the capacities in the circuit and the average value detected is not to be considered a reliable real value.
- ❑ The MCP command [SBCHS] indicates the percentage of charge and is not linear with respect to the battery voltage, however approximately it reads 0% with a voltage equal to about 3.2 V, while 100% occurs with about 4.1 V. These values are influenced by the temperature and the total operating time that worked the battery.

ELECTRICAL CONNECTION CONVERTER- SENSOR



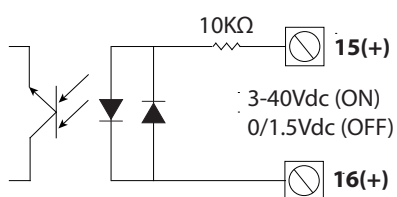
Sudden movements of the electrodes cable could introduce noise.

SH = SHIELD OF CABLE internally connected to ground.

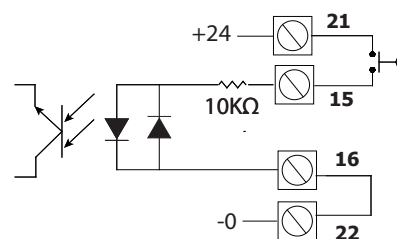


DIGITAL INPUT ON/OFF OPERATION

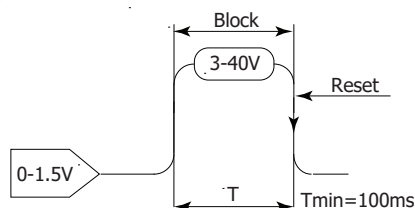
EXTERNAL POWER SUPPLY



INTERNAL POWER SUPPLY



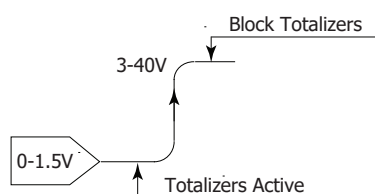
RESET TOTALIZERS



Necessary conditions for enabling the function

- ☐ POS. 6.1 see page 28 (T+; total direct positive set on)
- ☐ POS. 6.2 see page 28 (P+; partial direct positive set on)
- ☐ POS. 6.3 see page 28 (T-; total direct negative set on)
- ☐ POS. 6.4 see page 28 (P-; total direct negative set on)

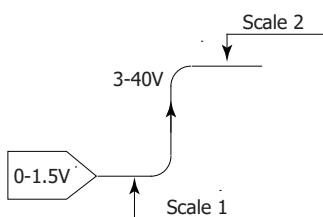
BLOCK TOTALIZERS



Necessary conditions for enabling the function

- ☐ POS. 6.5 see page 28 (Totalizer counting lock command set on)

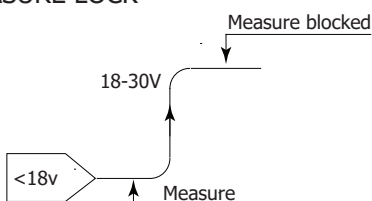
RANGE CHANGE



Necessary conditions for enabling the function

- ☐ POS. 6.8 see page 28 (Range change set on)

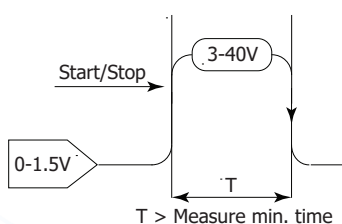
MEASURE LOCK



Necessary conditions for enabling the function

- ☐ POS. 6.6 see page 28 (Totalizer counting lock command set on)

BATCHING



Necessary conditions for enabling the function

- ☐ 7.1 Output 1 functions, Set to BATCHING

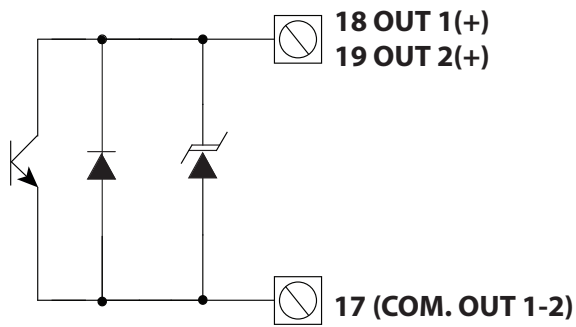


MUST BE
T > Tmin

MEASURE FREQ.	T MIN
50HZ	20ms
10HZ	100ms
5HZ	200ms
2HZ	500ms

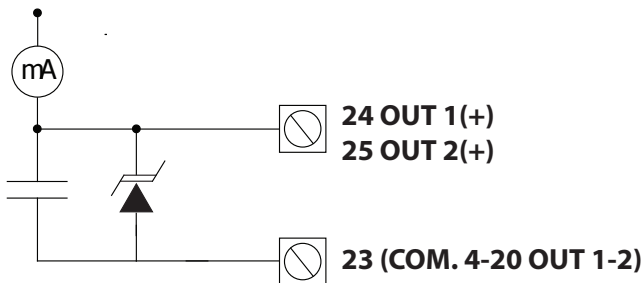
OUTPUTS WIRING

DIGITAL OUTPUTS



- ☐ Opto-insulated output with floating collector and emitter terminals freely connectable
- ☐ Maximum switching voltage: 40V ---
- ☐ Maximum switching current: 150mA
- ☐ Maximum saturation voltage between collector and emitter @150mA: 1.2V
- ☐ Maximum switching frequency (load on the collector or emitter, $R_L=470\Omega$, $V_{OUT}=24V$ ---): 1250Hz
- ☐ Maximum reverse current bearable on the input during and accidental polarity reversion (VEC): 100mA
- ☐ Insulation from other secondary circuits: 500V ---

ANALOG OUTPUTS



- ☐ Opto-insulated output
- ☐ Maximum load: 1000Ω
- ☐ Maximum voltage without load: 27V
- ☐ Refresh frequency is the same of the sample frequency of the connected sensor
- ☐ Protected against persistent over voltages to maximum 30V

RS485 MODBUS MODULE (OPTIONAL)

Positioning to 'ON' the termination switches 1 and 2, a 120Ω resistance is activated in the RS485 circuit (see terminal block).

